



Service Tool Instruction

Service Tool Instruction
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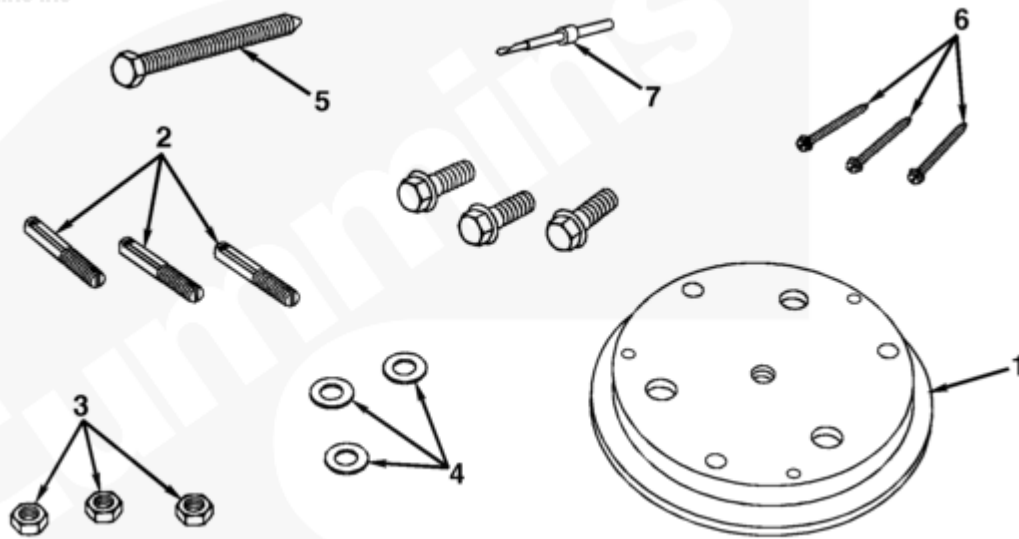
Description

Thick Wear Sleeve and Wet Flywheel Housing Oil Seal Installation and Removal

Purpose

This service tool instruction provides procedures for the thick wear sleeve and wet flywheel housing oil seal installation and removal tool, Part Number 3163826. This tool is used to install and remove the Signature, ISX, and QSX15 rear crankshaft thick wear sleeve and wet flywheel housing oil seal.

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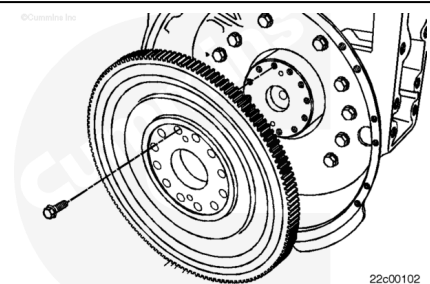
Table 1, Thick Wear Sleeve and Wet Flywheel Housing Oil Seal Installation and Removal Kit, Part Number 3163826

Item Number	Part Number	Description	Quantity
1	3163827	Thick Wear Sleeve/Wet Flywheel Housing Oil Seal Mandrel	1
2	3163706	Puller Jaws	3
3	S-217	Hexagon Nut (7/16-14)	3

Table 1, Thick Wear Sleeve and Wet Flywheel Housing Oil Seal Installation and Removal Kit, Part Number 3163826

Item Number	Part Number	Description	Quantity
4	3000082	Flat Washer (7/16 x 7/8)	3
5	3163057	Forcing Screw	1
6	3163828	Number 10-2 Sheet Metal Screw	3
7	3163089	Pilot Drill	1

Remove the flywheel from the engine. Refer to Procedure 016-005 (/qs3/pubsys2/xml/en/procedures/10/10-016-005.html) in the Troubleshooting and Repair Manual, Signature, ISX, and QSX15 Engines, Bulletin 3666239.



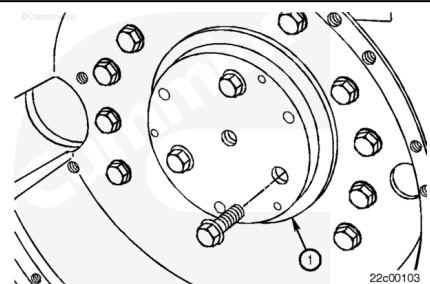
⚠ CAUTION ⚠

Be careful not to drop the thick wear sleeve and wet flywheel housing oil seal mandrel. Failure to do so could cause bodily injury and or damage to the mandrel tool.

Position the thick wear sleeve and wet flywheel housing oil seal mandrel (1) onto the crankshaft.

Install three flywheel mounting capscrews into the thick wear sleeve and wet flywheel housing oil seal mandrel.

Finger-tighten the capscrews.

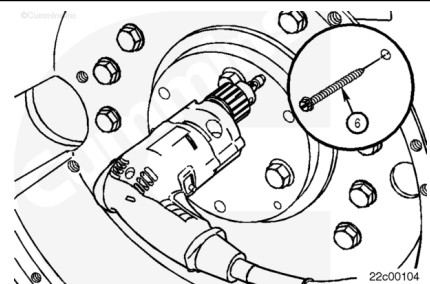


Do **not** overtighten the metal screws.

Insert the pilot drill into the chuck of drill and tighten the chuck.

Drill three holes into the wet flywheel housing oil seal.

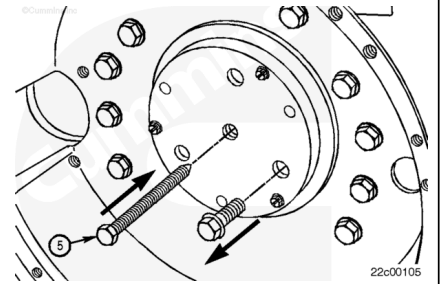
Install the three Number 10-2 sheet metal screws (6) into the drilled holes of the wet flywheel housing oil seal and hand-tighten.



Remove the three flywheel mounting capscrews.

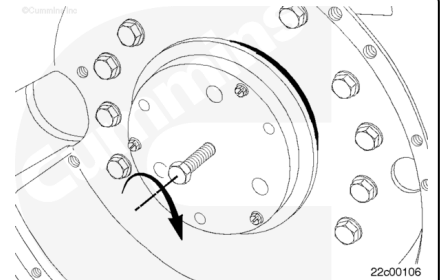
Lubricate the tip of the forcing screw (5).

Install the forcing screw (5) into the center of the thick wear sleeve and wet flywheel housing oil seal mandrel.



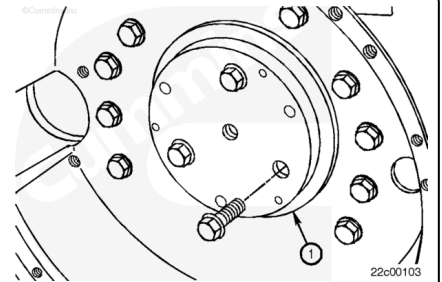
Continue to rotate the forcing screw **clockwise** until the oil seal is removed.

Remove the three 10-2 sheet metal screws and discard the used oil seal.



Position the thick wear sleeve/wet flywheel housing oil seal mandrel (1) onto the crankshaft.

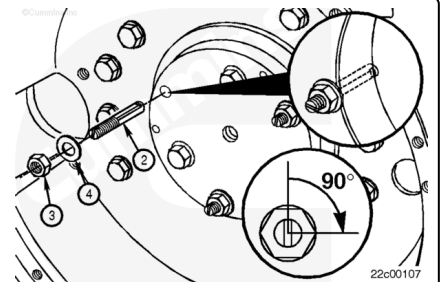
Install the three capscrews from the flywheel and hand-tighten.



Rotate the jaw pullers (2) until the large flat edge is aligned with the thick wear sleeve. Insert the jaw pullers until they bottom in the bore. After all three jaw pullers are inserted, rotate each one **clockwise** 90 degrees. This will correctly position the jaw pullers in relation to the thick wear sleeve.

Install the three flat washers (4) onto the jaw pullers.

Install the three hexagon nuts (3) onto the jaw pullers, and tighten.

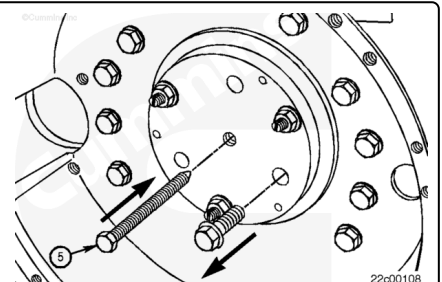


Torque Value: 11 n·m [97 in-lb]

Remove the three flywheel mounting capscrews.

Install the forcing screw (5) into the center of the thick wear sleeve and wet flywheel housing oil seal mandrel.

Rotate the forcing screw **clockwise** until the thick wear sleeve is removed.

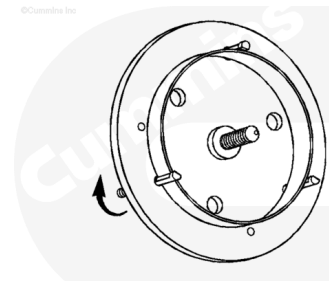


Remove the thick wear sleeve and wet flywheel housing oil seal mandrel and the thick wear sleeve assembly from the rear of the crankshaft. Loosen the nuts on the jaw puller.

Rotate the three jaw pullers **counterclockwise** 90 degrees to unlock the wear sleeve.

Remove the forcing screw.

Clean all the parts in the thick wear sleeve/wet flywheel housing oil seal installation and removal kit, and store in a safe place.



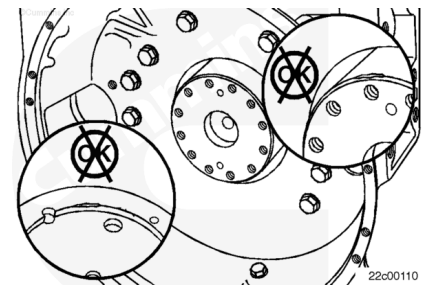
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Clean and inspect the crankshaft surface and the thick wear sleeve and wet flywheel housing oil seal mandrel.

Use crocus cloth to remove any dirt or deposits on both the tool and the crankshaft.

Use a lint-free cloth to remove any traces of oil from the crankshaft flange and the thick wear sleeve and wet flywheel housing oil seal mandrel.

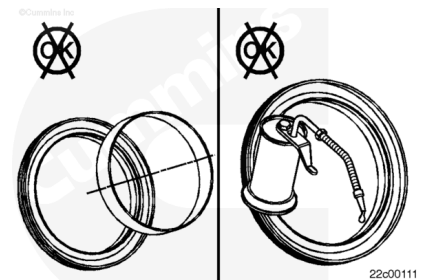
Check the crankshaft for nicks, burrs, and grooves.



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Do **not** separate the wet flywheel housing oil seal from the thick wear sleeve at any time during the assembly process. A new thick wear sleeve and seal assembly **must** be used.

Do **not** use any lubricant to install the wet flywheel housing oil seal. The lip of the oil seal and the crankshaft **must** be oil free and dry for proper installation of the wet flywheel housing oil seal.

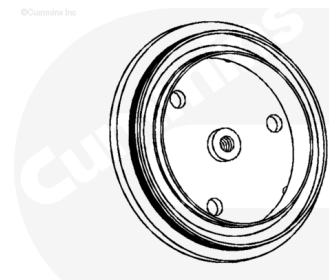


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Note : The yellow lip of the wet housing oil seal **must** be facing outward (toward the transmission).

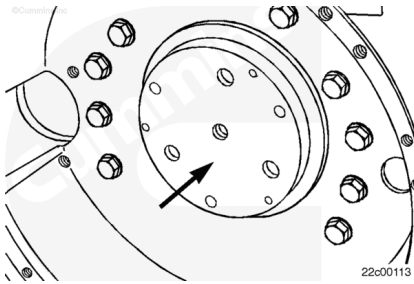
The thick wear sleeve and wet flywheel housing oil seal **must** be installed as one assembly.

Position the thick wear sleeve and the wet housing oil seal into the proper counterbore location in the thick wear sleeve and wet flywheel housing oil seal mandrel.



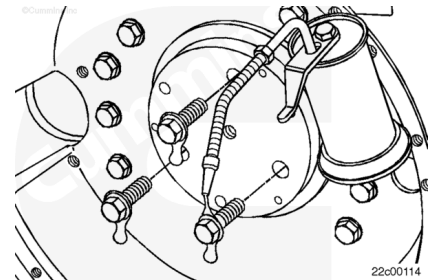
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Position the thick wear sleeve, wet housing oil seal, and thick wear sleeve and wet flywheel housing oil seal mandrel against the crankshaft flange.



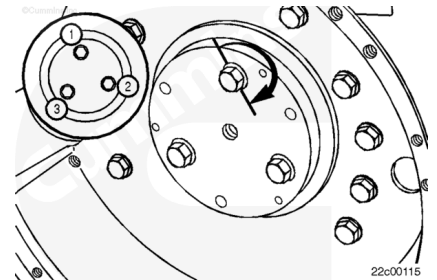
Lubricate the backside of the capscrew so that the lubricated side faces toward the thick wear sleeve and wet flywheel housing oil seal mandrel.

Install the three capscrews and finger-tighten.



⚠ CAUTION ⚠

When installing the wet flywheel housing oil seal and thick wear sleeve into the bore area, make certain that the thick wear sleeve remains parallel to the housing when tightening the capscrews. Failure to do so can result in damage to the wet housing oil seal and thick wear sleeve.



Tighten the capscrews 1/2 of a revolution at a time. Alternately tighten the capscrews in a **clockwise** pattern until the thick wear sleeve and wet flywheel housing oil seal mandrel seats against the crankshaft flange as shown in the illustration.

Torque Value: 68 n•m [50 ft-lb]

Remove the three capscrews.

Remove the mandrel.

Install the flywheel housing. Refer to Procedure 016-005 (</qs3/pubsys2/xml/en/procedures/10/10-016-005.html>) in the Troubleshooting and Repair Manual, Signature, ISX, and QSX15 Engines, Bulletin 3666239, for the capscrew torque information.

Clean all the parts in the thick wear sleeve and wet flywheel housing oil seal installation and removal tool kit, and store in a safe place.

